3.8.26

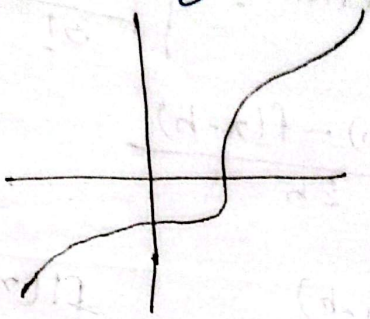
Ch-4
Root finding

$f(x) = 0$

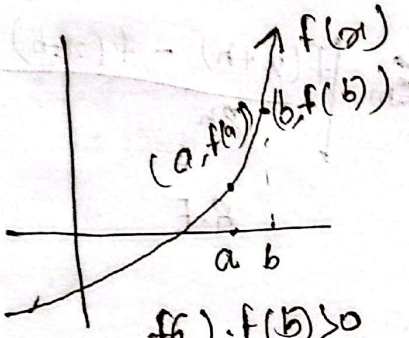
$e^x \sin 2x = 0$

$x^2 - 5x + 6 = 0$

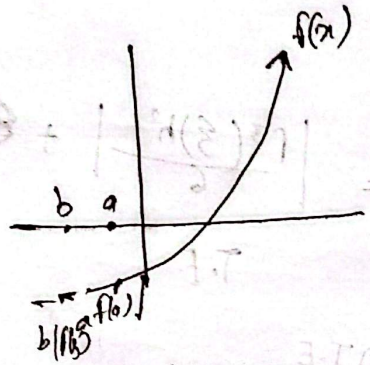
$x + 2 = 0$



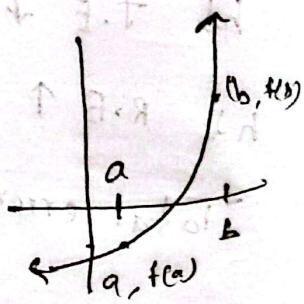
Bisection Method:



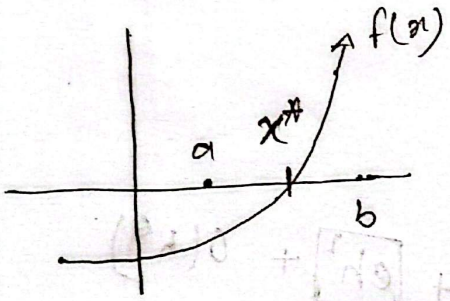
$f(a) \cdot f(b) > 0$



$f(b) \cdot f(a) > 0$



$f(a) \cdot f(b) < 0$



Ex: $f(x) = x^8 - 7x^2 + 14x - 6$ $[1, 3.2]$ $\epsilon = 10^{-3}$

$f(x) = 0$

k	a_k	m_k	b_k	$f(a_k)$	$f(m_k)$	$f(b_k)$
0	1	$\frac{1+3.2}{2} = 2.1$	3.2	2	1.791	−0.112
1	2.1	2.65	3.2	1.791	0.552	−0.112
2	2.65	2.925	3.2	0.552	0.0858	−0.112
3	2.925	3.0625	3.2	0.0858	−0.054	−0.112
4	2.925	2.9937	3.0625	0.0858	0.006	−0.054

$|f(m_k)| \leq \epsilon$

9 2.998047 3.000195 3.0023 0.0019 −0.000195 −0.00233

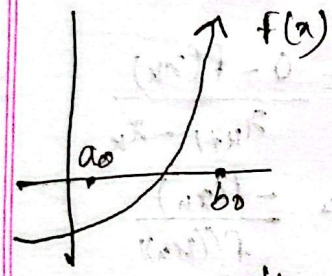
$m_k = 3.000195$

number of iteration, $n \geq \dfrac{\log|b_0 - a_0| - \log\epsilon}{\log 2} - 1$

$[a_0, b_0]$, ϵ

$$n \geq \frac{\log|3.2 - 1| - \log 10^{-3}}{\log 2} - 1$$

$n \geq 10.10 \approx 10$

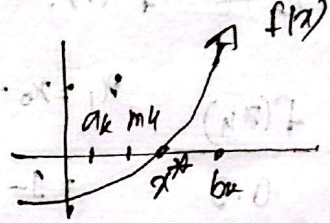


$k = 0 \quad |b_0 - a_0|$

$k = 1 \quad \dfrac{|b_0 - a_0|}{2}$

$k = 2 \quad \dfrac{|b_0 - a_0|}{2^2}$

$k \quad \dfrac{|b_0 - a_0|}{2^k} = \dfrac{|b_k - a_k|}{2}$



$\epsilon, [a_0, b_0]$

$|m_k - x^*| \leq \dfrac{|b_k - a_k|}{2}$

$|m_k - x^*| \leq \dfrac{|b_k - a_k|}{2^k} \Big/ 2$

$|m_k - x^*| \leq \dfrac{(b_0 - a_0)}{2^{k+1}}$

$\dfrac{|b_0 - a_0|}{2^{k+1}} \leq \epsilon$

$\log \dfrac{|b_0 - a_0|}{2^{k+1}} \leq \log\epsilon$

$\Rightarrow \log|b_0 - a_0| - \log(2^{k+1}) \leq \log\epsilon$

$\Rightarrow \log|b_0 - a_0| - \log\epsilon \leq (k+1)\log 2$

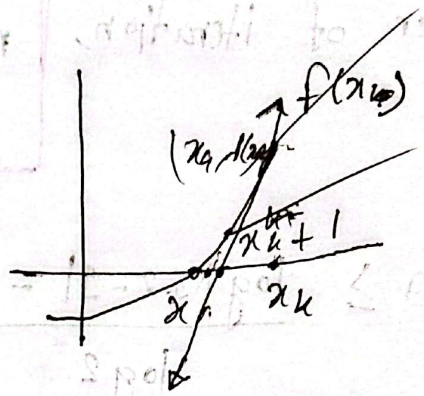
$\Rightarrow \dfrac{\log|b_0 - a_0| - \log\epsilon}{\log 2} \leq k + 1$

$\Rightarrow \dfrac{\log|b_0 - a_0| - \log\epsilon}{\log 2} - 1 \leq k$

$$\therefore k \geq \frac{\log|b_0 - a_0| - \log\epsilon}{\log 2} - 1$$

Newton raphson method:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$



Ex: $f(x) = \frac{1}{x} - 0.5$, $x_0 = 1$

$\epsilon = 10^{-5}$ $\quad f'(x) = -\frac{1}{x^2}$

k	x_k	$f(x_k)$
0	1	0.5
1	1.5	0.167
⋮		
6	2.000	0.0000

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$= 1 - \frac{f(1)}{f'(1)}$

$= 1 - \frac{0.5}{-1}$

slope $= \frac{y_2 - y_1}{x_2 - x_1}$

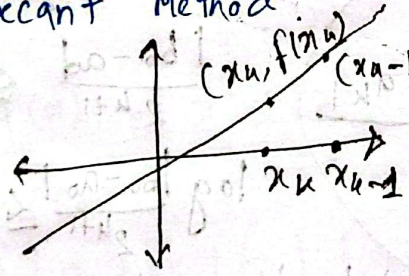
$f'(x_k) = \frac{0 - f(x_k)}{x_{k+1} - x_k}$

$\Rightarrow x_{k+1} - x_k = \frac{-f(x_k)}{f'(x_k)}$

$\therefore x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$

Secant Method:

initial condition → 2



$f'(x) = \frac{f(x_{k-1}) - f(x-h)}{x - (x-h)}$

$f'(x_k) = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$

$$x_{k+1} = x_k - \frac{f(x_k)}{\frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}}$$

$$x_{k+1} = x_k - \frac{f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$$

10.8.26

Fixed point iterations

$f(x) = 0$

$\rightarrow g(x) = x$

x_0

$x_{n+1} = g(x_n)$

$f(x) = x^2 - 2x - 3 = 0$

$x^2 - 2x - 3 = 0$

$\Rightarrow x^2 = 2x + 3$

$\therefore x = \boxed{\sqrt{2x+3}}$

$g(x)$

$x_0 = 0$

$g(0) = \sqrt{2.073} = 1.7321$

$g(1.732) \doteq 2.5495$

$g(2.5495) = 2.843$

$g(2.843) \approx 2.9471$

$g(2.9471) = 2.9822$

$g(2.9822) = 2.9940$

$g(2.9940) = 2.9980$

$g(2.9980) \doteq 3$

$g(3) = 3$

$\therefore x^* = 3$

Ans

$x^* = 3$ converge

$x^2 - 2x - 3 = 0$

$\Rightarrow x^2 - x - x - 3 = 0$

$\Rightarrow x = \boxed{x^2 - x - 3}$

$g(x)$

$x_0 = 0$

$g(0) = -3$

$g(-3) = 9$

$g(9) = 69$

$g(69) = 4.69 \times 10^3$

$\vdots$

diverges

no solution.

$x^2 - 2x - 3 = 0$

$\Rightarrow 2x^2 - x^2 - 2x - 3 = 0$

$\Rightarrow 2x^2 - 2x = x^2 + 3$

$\Rightarrow x(2x-2) = x^2 + 3$

$\Rightarrow x = \underbrace{\boxed{\frac{x^2+3}{2x-2}}}_{g(x)}$

$x_0 = 0$

$g(0) = -1.5$

$g(-1.5) = -1.05$

$g(-1.05) = -1$

$g(-1) = -1$

$\therefore x^* = -1$

if $x_0 = 42$

$g(42) = 9.33$

$g(9.33) = 4.65$

$g(4.65) = 3.51$

$g(3.51) = 3.12$

3.06

3.02

3.01

3

$x_0 = 42$

42

↓

1722

↓

2.95×10^6

↓

⋮

$x_0 = 42$

$g(42) = 21.6$

$g(21.6) = 11.4$

$g(11.4) = 6.39$

↓

$g(6.39) = 4.07$

$g(4.07) = 3.19$

$g(3.19) = 3.01$

$g(3.01) = 3$

$g(3) = 3$

$x^* = 3$

Contraction mapping theorem:

$g(x)$ converges to x^* if and only if

$$\lambda = |g'(x^*)| < 1$$

Ex: $f(x) = x^2 - 2x - 3 = 0$

$$x^* = -1, 3$$

i) $g(x) = \sqrt{2x+3}$

$$g'(x) = \frac{2}{2\sqrt{2x+3}} = \frac{1}{\sqrt{2x+3}}$$

$$\lambda_1 = |g'(-1)| = 1$$

$$\lambda_2 = |g'(3)| = \frac{1}{3}$$

$g(x)$ converges to 3

$\lambda_2 < 1$

ii) $g(x) = x^2 - x - 3$

$$g'(x) = 2x - 1$$

$$\lambda_1 = |g'(-1)| = |-3| = 3$$

$$\lambda_2 = |g'(3)| = 5$$

$\lambda_1, \lambda_2 > 1$, diverge

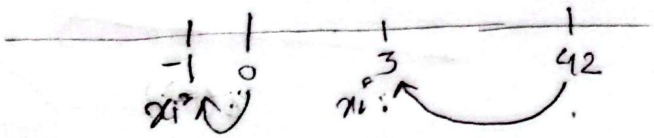
iii) $g(x) = \frac{x^2+3}{2x-2}$

$$g'(x) = \frac{(2x-2)\cdot 2x - (x^2+3)\cdot 2}{(2x-2)^2}$$

$$= \frac{4x^2 - 4x - 2x^2 + 6}{(2x-2)^2} = \frac{2x^2 - 4x - 6}{(2x-2)^2}$$

$\lambda_1 = g'(-1) = 0$ $\quad \therefore |\lambda_1|, |\lambda_2| < 1$

$\lambda_2 = g'(3) = 0$ $\quad$ converge (fast)



$\lambda = 0$ super linearly convergent

$0 < \lambda < 1$ linearly convergent

$\lambda \geq 1$ divergent

Prove that Newton-Raphson's method is super linearly convergent

$$x_{k+1} = \underbrace{x_k - \frac{f(x_k)}{f'(x_k)}}_{g(x)}$$

$\lambda = 0 \Rightarrow |g'(x^*)| = 0$

$$g(x) = x - \frac{f(x)}{f'(x)}$$

let's consider x^* is a solution of $f(x)$

$$g'(x) = 1 - \frac{f'(x)\frac{d}{dx}(f(x)) - f(x)\cdot\frac{d}{dx}(f'(x))}{\{f'(x)\}^2}$$

$$= 1 - \frac{\{f'(x)\}^2 - f(x)\cdot f''(x)}{\{f'(x)\}^2}$$

$$= \frac{\{f'(x)\}^2 - \{f'(x)\}^2 + f(x)\cdot f''(x)}{\{f'(x)\}^2}$$

$$= \frac{f(x)\, f''(x)}{\{f'(x)\}^2}$$

$$|g'(x^*)| = \left| \frac{f(x^*)\, f''(x^*)}{\{f'(x^*)\}^2} \right| = 0 \quad \therefore \lambda = 0$$